

Disk Analyzer

Information security

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Introduction

DiskAnalyzer is a Python-based tool designed to scan directories, calculate folder sizes, and generate detailed disk usage reports. The program helps users understand how disk space is distributed across the file system by identifying large directories and files.

The tool generates reports in text format and also supports optional automatic email delivery of the results. This makes it useful for regular system monitoring

Features

- Recursively scans folders and subfolders
- Calculates total directory sizes
- Generates text reports
- Optional JSON export
- Email report support
- Handles permission errors safely
- Works on macOS, Linux, and Unix-like systems



Installation

Clone the repository:

```
git clone https://github.com/khatszianAli/information_security.git  
cd information_security/final_project/DiskAnalyzer
```



Make the script executable (optional):

```
chmod +x DiskAnalyzer.py
```



Check your Python version:

```
python3 --version
```



Usage Basic command:

```
python3 DiskAnalyzer.py --root /path/to/directory --out report.txt --json
```



macOS example:

```
python3 DiskAnalyzer.py --root /Users/yourusername --out ~/disk_reports/disk_report_$(date +%F).txt --json
```



Installation

Email Reports To enable sending reports by email, set these environment variables:

```
export EMAIL_USER="your_email@gmail.com"  
export EMAIL_PASS="your_app_password"  
export EMAIL_TO="receiver@email.com"  
export SMTP_SERVER="smtp.gmail.com"  
export SMTP_PORT="587"
```



Run with email enabled:

```
python3 DiskAnalyzer.py --root /Users/yourusername --out report.txt --json --mail
```



Automatization

Automate with cron (daily run) Example cron job (runs every day at 21:00):

```
00 21 * * * /opt/homebrew/bin/python3 /path/to/DiskAnalyzer.py \  
--root /Users/yourusername \  
--out /Users/yourusername/disk_reports/disk_report_$(date +%F).txt \  
--json --mail >> /Users/yourusername/disk_reports/disk_analyzer.log 2>&1
```

The report

Disk Analyzer Report

Generated: 2025-12-04 21:08:48

Root scanned: /Users/newowner

Total size under root: 121.95 GB

Top 30 largest directories:

1. 121.95 GB /Users/newowner
2. 78.54 GB /Users/newowner/Library
3. 27.32 GB /Users/newowner/Library/Application Support
4. 20.00 GB /Users/newowner/Library/Developer
5. 17.16 GB /Users/newowner/.conan2
6. 17.16 GB /Users/newowner/.conan2/p
7. 14.39 GB /Users/newowner/Library/Application Support/Steam
8. 12.95 GB /Users/newowner/Library/Application Support/Steam/steamapps
9. 12.70 GB /Users/newowner/.conan2/p/b
10. 12.57 GB /Users/newowner/Library/Application Support/Steam/steamapps/downloading
11. 12.57 GB /Users/newowner/Library/Application Support/Steam/steamapps/downloading/444200
12. 12.57 GB /Users/newowner/Library/Application Support/Steam/steamapps/downloading/444200/World of Tanks Blitz.app
13. 12.57 GB /Users/newowner/Library/Application Support/Steam/steamapps/downloading/444200/World of Tanks Blitz.app/Contents
14. 12.54 GB /Users/newowner/Library/Developer/CoreSimulator
15. 12.34 GB /Users/newowner/Library/Application Support/Steam/steamapps/downloading/444200/World of Tanks Blitz.app/Contents/Resources
16. 12.34 GB /Users/newowner/Library/Application Support/Steam/steamapps/downloading/444200/World of Tanks Blitz.app/Contents/Resources/Data
17. 11.97 GB /Users/newowner/Library/Application Support/Steam/steamapps/downloading/444200/World of Tanks Blitz.app/Contents/Resources/Data/3d
18. 9.85 GB /Users/newowner/Library/Application Support/Steam/steamapps/downloading/444200/World of Tanks Blitz.app/Contents/Resources/Data/3d/Tanks
19. 9.50 GB /Users/newowner/Library/Group Containers
20. 8.69 GB /Users/newowner/operating_system
21. 8.69 GB /Users/newowner/operating_system/projects
22. 8.69 GB /Users/newowner/operating_system/projects/project-1-khatszianAli
23. 8.69 GB /Users/newowner/operating_system/projects/project-1-khatszianAli/platforms
24. 8.52 GB /Users/newowner/Library/Group Containers/6N38VWS5BX.ru.keepcoder.Telegram
25. 8.52 GB /Users/newowner/Library/Group Containers/6N38VWS5BX.ru.keepcoder.Telegram/appstore
26. 8.46 GB /Users/newowner/Library/Group Containers/6N38VWS5BX.ru.keepcoder.Telegram/appstore/account-13988570551400134924
27. 8.27 GB /Users/newowner/Library/Group Containers/6N38VWS5BX.ru.keepcoder.Telegram/appstore/account-13988570551400134924/postbox
28. 7.95 GB /Users/newowner/Library/Group Containers/6N38VWS5BX.ru.keepcoder.Telegram/appstore/account-13988570551400134924/postbox/media
29. 7.60 GB /Users/newowner/Library/Caches
30. 7.44 GB /Users/newowner/Library/Developer/Xcode

Notes:

- Sizes are recursive (include subdirectories).
- Some files/dirs may be skipped if permission denied.

(Report attached.)

Conclusion

In conclusion, the DiskAnalyzer project demonstrates how Python can be used to effectively monitor and analyze disk usage in a file system. The tool successfully scans directories, calculates their sizes, and generates clear and informative reports that help users identify storage issues and manage disk space more efficiently. By supporting report export and optional email notifications, DiskAnalyzer can be integrated into automated system administration tasks. This project also strengthens practical skills in file system operations, scripting, and basic security awareness.

Thank You